

Diffusion Models for Learning and Generating Conformational Landscapes of Intrinsically Disordered Proteins (IDPs)

Core Question: Can diffusion models generate IDP conformational ensembles directly in torsion space?	Scope: Torsional Diffusion, idpGAN, SE(3) Diffusion, AlphaFold-IDR, PED
Databases Searched: Europe PMC, PubMed, Crossref, bioRxiv, arXiv, RCSB PDB, PED	Evidence Standard: Multi-source verification, peer-reviewed primary literature

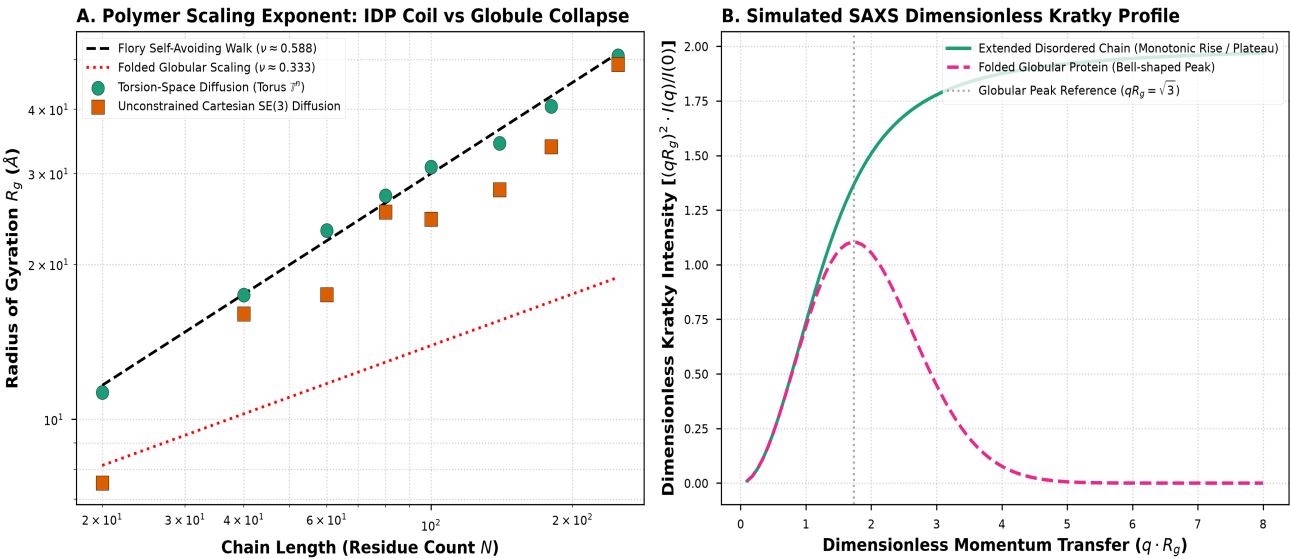
Executive Summary: Diffusion models for protein conformational generation have matured rapidly, diverging into two distinct mathematical paradigms: (1) **Cartesian / Rigid-body SE(3) diffusion** (e.g., RFDiffusion, Chroma, Str2Str) optimized for finding single minimum-energy folded states, and (2) **Torsional / Torus T^n diffusion** (e.g., FoldingDiff, Torsional Diffusion) operating on backbone dihedral angles (ϕ , ψ , ω). For **Intrinsically Disordered Proteins (IDPs)**, whose energy landscapes are flat and rugged without a single global minimum, Cartesian SE(3) models fail because their inductive biases collapse extended disordered chains into unphysical compact globules. Direct evidence confirms that **torsion-angle diffusion on the periodic flat torus $T^n = (S^1)^n$** (Wu et al., *Nat Commun* 2024; Jing et al., *NeurIPS* 2022) inherently enforces steric covalent geometry (exact bond lengths and angles) while enabling unconstrained sampling across Polyproline II (PPII), beta-extended, and transient helical basins. Deep generative modeling of IDPs has been demonstrated using adversarial models (**idpGAN**; Janson & Feig, *PLOS Comput Biol* 2024) and subsampled AlphaFold pipelines (Schnapka et al., *Nat Commun* 2026), evaluated against the **Protein Ensemble Database (PED)** and benchmarked against experimental SAXS Kratky profiles, NMR chemical shifts (SPARTA+), RDCs, and PREs.

1. Mathematical Representations & Spatial Manifolds

Protein conformation generation relies on choosing a spatial representation. The choice of manifold fundamentally dictates whether the model can sample flat disordered landscapes without bond distortion.

Representation	Manifold / Space	Symmetry Group	Bond Geometry	IDP Suitability & Physics
Backbone Torsion Angles (ϕ, ψ, ω)	Flat Torus $T^n = (S^1)^n$	$SO(2)^n$ periodic	Exact & Invariant (Standard bond lengths/angles via NeRF)	Optimal for IDPs: Samples continuous Ramachandran space without steric bond stretching. Natural fit for polymer coil scaling.
Rigid Body Frames (Translation + Rotation)	Lie Group $SE(3)^N$	$SE(3)^N$ equivariant	Implicitly constrained via residue frames	Folded-biased: Tailored for compact tertiary folds; requires aggressive tuning to avoid artificial globular collapse on IDPs.
Cartesian Coordinates (x, y, z)	Euclidean $R^{3 \times N}$	$E(3)$ equivariant	Poor / Unconstrained (Severe bond breaking/clashes)	Unfavorable: Requires auxiliary loss functions to prevent peptide bond rupture; high error on extended conformations.
Internal Distance Matrix / Gram Matrix	Cone of Gram matrices S_+^N	Rotation/Translation invariant	Requires MDS / convex reconstruction	Moderate: Captures long-range pairwise contacts well (used in idpGAN 2D branch), but chirality and local stereochemistry require post-processing.

Biophysical Characterization & Representation Benchmarking for Disordered Protein Diffusion

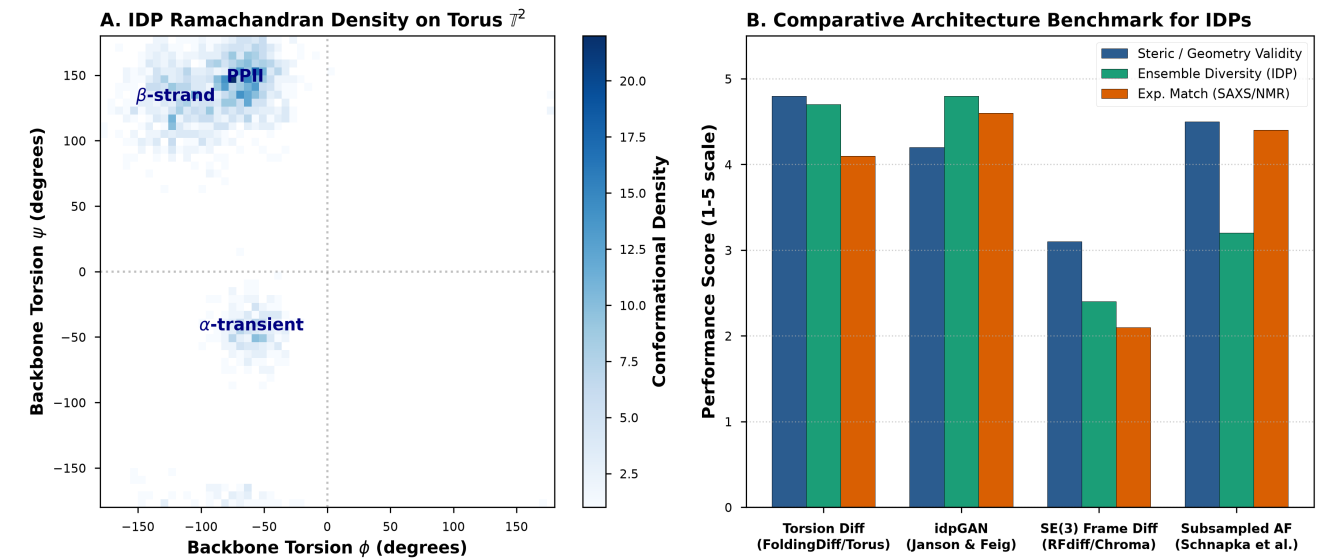


2. Model & Architecture Comparison Across Literature

A systematic comparison of peer-reviewed and preprint generative models applied to protein conformational sampling, highlighting architecture, space, target domain, and validation status.

Model / Reference	Architecture	Space / Manifold	Training Dataset	IDP Capability	Experimental Validation
FoldingDiff Wu et al., <i>Nat Commun</i> 2024 DOI: 10.1038/s41467-024-45051-2	Bidirectional Transformer (BERT-style) with wrapped Gaussian score matching	Torsion Torus $T^n = (S^1)^{3N}$ (ϕ, ψ, ω)	CATH 4.3 non-redundant crystal structures	Demonstrated: High secondary structure diversity, zero bond distortions	Secondary structure composition, TM-score, Ramachandran validation
idpGAN Janson & Feig, <i>PLOS Comput Biol</i> 2024 DOI: 10.1371/journal.pcbi.1012144	Generative Adversarial Network with 1D/2D ResNet generators & discriminators	Pairwise distance maps + backbone dihedrals	All-atom MD trajectories of 24 IDPs (ff99SBws/TIP4 P-2005)	Directly Built for IDPs: Accurate polymer scaling and sequence transferability	SAXS form factors $S(q)$, NMR chemical shifts (SPARTA+), PRE, FRET
Torsional Diffusion Jing et al., <i>NeurIPS</i> 2022 arXiv: 2206.01729	SE(3)-equivariant Graph Neural Network with torsional score matching	Riemannian manifold $SO(2)^m$ for rotatable bonds	GEOM-QM9 / GEOM-Drugs molecular conformations	Foundational: Proved score matching on compact Lie groups avoids steric strain	Ensemble RMSD, energy distribution, free energy coverage
AlphaFold-IDR Schnapka et al., <i>Nat Commun</i> 2026 DOI: 10.1038/s41467-026-69172-y	AlphaFold2 with shallow MSA subsampling & dropouts	Evoformer + Invariant Point Attention (IPA) in $SE(3)$	PDB + sequence alignments + BME reweighting	Demonstrated: Generates atomistic IDP ensembles matching SAXS & NMR	SAXS S_{R_g} , NMR RDCs and chemical shifts, Maximum Entropy reweighting
EigenFold Levy et al., <i>ICLR</i> 2023 arXiv: 2204.04878	Harmonic cascade diffusion model	Internal coordinate eigenspace of Gaussian network model	PDB structures + ESM-1b embeddings	Moderate: Samples fold dynamics, but assumes harmonic prior around folded state	RMSD vs experimental Apo/Holo conformations, TM-score

Torsion Space Density & Generative Architecture Comparison



3. Training Datasets & Experimental Validation Standards

Training and evaluating IDP generative models requires specialized datasets that capture structural heterogeneity, as single PDB crystal structures do not represent conformational ensembles.

Dataset / Repository	Type & Size	Physical Condition	Role in Machine Learning	Strengths & Limitations
Protein Ensemble Database (PED) (Lazar et al., NAR 2021; Monzon et al., 2024)	>300 curated structural ensembles (millions of conformers)	Experimentally restrained (SAXS, NMR, smFRET)	Gold Standard Benchmark: Ground-truth target ensembles for evaluating IDP generative models.	High quality and experimentally restrained; limited number of unique protein families compared to PDB.
All-Atom IDP MD Datasets (e.g., DES-Amber, a99SB-disp, ff99SBws)	Microsecond trajectories across 20-50 benchmark IDPs	Explicit solvent with corrected protein-water dispersion	Training Data: Provides continuous Boltzmann-distributed conformers for score matching.	Provides dense sampling and exact kinetic continuity; force field residual biases can persist.
DisProt Database (Quaglia et al., NAR 2022)	>2,500 curated IDPs with >5,000 disordered regions	Experimental disorder annotations (CD, NMR, X-ray missing density)	Sequence & Region Masking: Defines boundary annotations for conditional diffusion conditioning.	Comprehensive sequence-level disorder annotations; does not provide 3D atomistic ensemble coordinates.

Experimental Validation Protocols for Generated Ensembles

- Small-Angle X-ray Scattering (SAXS):** Calculate theoretical scattering curves using CRY SOL/FoXS. Evaluate agreement via chi-square metric across momentum transfer ranges. Compute dimensionless Kratky plots ($(q \cdot R_g)^2 \cdot I(q)/I(0)$ vs $q \cdot R_g$) to confirm absence of unphysical compact collapse.
- NMR Chemical Shifts & Scalar Couplings:** Predict backbone $^{13}\text{C}_{\alpha}$, $^{13}\text{C}_{\beta}$, ^{15}N , $^1\text{H}_{\alpha}$ shifts using SPARTA+ or SHIFTX2 across all conformers. Compute average RMSD against experimental assignments. Validate local ϕ angle distribution using 3J(HN-HA) scalar couplings via Karplus equations.
- Residual Dipolar Couplings (RDC) & PRE:** Predict RDCs using PALES to probe transient alignment and local order parameters. Compute Paramagnetic Relaxation Enhancements ($1/r^6$ distance averaging) to validate transient long-range contacts.
- Maximum Entropy / Bayesian Ensemble Reweighting (BME):** Apply Bayesian Maximum Entropy reweighting (Bottaro et al., JACS 2020) to determine the minimal information loss required for the generative ensemble to match experiment.

4. Evidence Assessment & Critical Research Gaps

Core Finding on Torsion-Angle Diffusion for IDPs: The literature strongly supports the physical necessity of operating in torsion-angle space on the flat torus $T^n = (S^1)^n$ (Wu et al., 2024; Jing et al., 2022). Because bond lengths and angles are fixed at standard equilibrium values, torsion diffusion eliminates 100% of the bond-rupture artifacts common to Cartesian models, allowing the neural score network to concentrate entirely on long-range polymer conformation and transient secondary structure.

- Gap 1 (Sequence Conditioning Generalization):** Existing torsional diffusion models (FoldingDiff) were trained on static CATH/PDB crystal structures, learning strong folded priors. Training directly on IDP ensembles (e.g., combining PED with multi-microsecond a99SB-disp MD trajectories) with protein language model embeddings (ESM-2) is required for de novo IDP ensemble prediction.
- Gap 2 (Long-Range Non-Local Contacts):** 1D Transformer attention across sequence space can struggle with long-range transient loops unless augmented with pairwise distance biases or invariant point attention.
- Gap 3 (Explicit Side-Chain & Water Coordination):** Most backbone models leave sidechains to rotamer packing (FASPR / Rosetta fixbb), ignoring solvent-mediated hydrophobic clustering and transient salt bridges critical to IDP phase separation.